Gravitational Potential Energy

Exam code: 0625 & 0972

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Gravitational potential energy

Extended tier only

- Energy in an object's gravitational potential energy store is defined as:

The energy an object has due to its height in a gravitational field

- **Work** is done against the **weight** force exerted on the object; therefore, energy is transferred
- This means that:
  - if an object is **lifted** up, energy will be transferred **to** its gravitational potential store
  - if an object is **lowered**, energy will be transferred **away from** its gravitational potential store

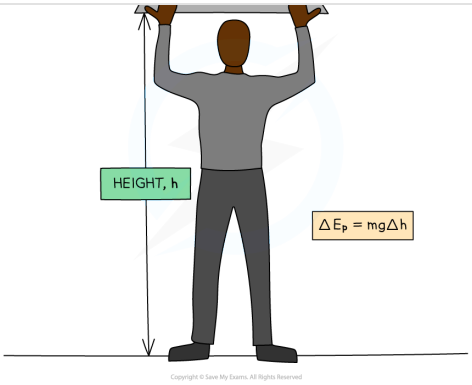
Gravitational potential energy of an object lifted through a gravitational field

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Energy is transferred to the mass's gravitational store as it is lifted through the gravitational field

Gravitational potential energy equation

- The change in energy in an object's gravitational potential energy store can be calculated using the equation:

$$\Delta E_p = mg\Delta h$$

- Where:
  - ΔE_p = change in gravitational potential energy, measured in joules (J)
  - m = mass, measured in kilograms (kg)
  - g = gravitational field strength, measured in newtons per kilogram (N/kg)
  - Δh = change in height, measured in metres (m)

Worked Example

A man climbs a flight of stairs that is, in total, 3.0 m higher than the floor. The man has a mass of 72 kg, and the gravitational field strength on Earth is approximately 9.8 N/kg.

Calculate the energy transferred to the man's gravitational potential energy store.

Answer:

Step 1: List the known quantities

- Mass of the man, $m = 72\ \text{kg}$
- Gravitational field strength, $g = 9.8\ \text{N/kg}$
- Change in height, $\Delta h = 3.0\ \text{m}$

Step 2: Write down the equation for gravitational potential energy

$$\Delta E_p = mg\Delta h$$

Step 3: Calculate the gravitational potential energy

Step 4: Round to an appropriate number of significant figures

- The input values are to 2 s.f. therefore, the output value must be to 2 s.f.

$$\Delta E_p = 2100 \text{ J } (2 \text{ s.f.})$$

Examiner Tips and Tricks

In your IGCSE exam, you will generally be expected to round your answers to an appropriate number of significant figures.

Your answer can only be as accurate as your least accurate input value. Therefore, your final answer should round to the lowest number of significant figures of all your input values.

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